UNIVERSITY OF CALIFORNIA AT BERKELEY

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SAN FRANCISCO • SANTA BARBARA • SANTA CRUZ

CONSENT TO PARTICIPATE IN RESEARCH
Uncovering the foundation of mathematical and probabilistic thought in a non-WEIRD culture

U.S. Adult Participants (online)

Key Information

- You are being invited to participate in a research study. Participation in research is completely voluntary.
- The purpose of the study is to investigate the foundations of mathematical and probabilistic thought and how these skills develop in different cultures.
- The study will take up to one hour and you will be asked to answer word problems about math and reasoning and/or make judgments about probability.
- Risks and/or discomforts may include the risk of breach of confidentiality.
- There is no direct benefit to you. The results from the study may allow educators to create interventions to help children develop math competency in the future.

Introduction

Steve Piantadosi and Fei Xu are professors in the Department of Psychology at the University of California, Berkeley. We are planning to conduct a research study, which I invite you to take part in.

You are being invited to participate in this study because you are seeking out research opportunities through Prolific.

Purpose

The purpose of this study is to investigate the cognitive foundations of mathematics and probabilistic thought and how these skills develop across cultures. The study is being conducted in small villages near San Borja, Bolivia as well as in your community in the United States in order to compare the development of mathematical thought across these two groups.

Procedures

If you agree to be in this study, you will be asked to complete the attached online study. You will be presented with one or more of the following tasks:

- **Math and reasoning word problems**: We will ask you to answer questions involving addition, subtraction, multiplication, division, and/or reasoning about probabilities. We will also ask you to explain the process you used to solve each problem.

- **Judgements about probability:** We will present you with boxes that contain objects of varying colors and/or shapes and ask you to predict which objects will be drawn randomly from the boxes. We may also present you with scenarios where a character draws out specific objects intentionally, and we will ask you questions about the character's goal.

Study time: Your study participation will take up to one hour total.

We may contact you via Prolific message or Prolific email to ask for clarification after you complete a task.

Benefits

There will be no direct benefit to you from participating in this study. However, we hope that the results of the research will allow educators to create interventions to help children develop math competency in the future.

Risks/Discomforts

- **Breach of confidentiality:** As with all research, there is a chance that confidentiality could be compromised; however, we are taking precautions to minimize this risk.

Confidentiality

- Your study data will be handled as confidentially as possible. If results of this study are published or presented, individual names and other personally identifiable information will not be used.

- To minimize the risks to confidentiality, we will not collect your name, email, or IP address in our survey questions. However, the server hosting the experiment may temporarily store your IP address in server logs. We will collect your Prolific ID along with the data you provide in the study. This identifying information will be stored in an encrypted file on secure, password-protected computers that belong to members of the study team. Identifying information will be destroyed 3 years after study completion. It is unlikely, but possible, that your data could accidentally be revealed.

- Your personal information may be released if required by law. Authorized representatives from the following organizations may review your research data for purposes such as monitoring or managing the conduct of this study:
  - National Science Foundation (NSF)
  - University of California

- ***Retaining research records:*** When the research is completed, we may save the study records for use in future research done by Steven Piantadosi, Fei Xu, or other researchers. We will retain de-identified study information indefinitely. The same measures described above will be taken to protect confidentiality of this study data.

- Identifiers might be removed from the identifiable private information. After such removal, the information could be used for future research studies or distributed to other investigators for future research studies without additional informed consent from the subject or the legally authorized representative.

Compensation

To thank you for participating in this study, you will receive compensation at a rate of $10 per hour. The exact amount of payment you receive will be calculated based on the time required to

respond to the task (e.g., if the task takes a full hour, you will receive $10; if it takes only 30 minutes, you will receive $5). You will receive this payment via credits through the Prolific system. No direct payments will be sent from the UC Berkeley research lab.

Rights

Participation in research is completely voluntary. You have the right to decline to participate or to withdraw at any point in this study without penalty or loss of benefits to which you are otherwise entitled.

Questions

If you have any questions or concerns about this study, you may contact Steven Piantadosi at stp@berkeley.edu or Fei Xu at fei_xu@berkeley.edu.

If you have any questions or concerns about your rights and treatment as a research subject, you may contact the office of UC Berkeley's Committee for the Protection of Human Subjects, at 510-642-7461 or subjects@berkeley.edu.

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If you agree to take part in the research, please print a copy of this page to keep for future reference, then click on the "Accept" button below.